Integration

COURSE ON SCRNA-SEQ DATA ANALYSIS

Putting datasets side by side without letting the batch decide the answer

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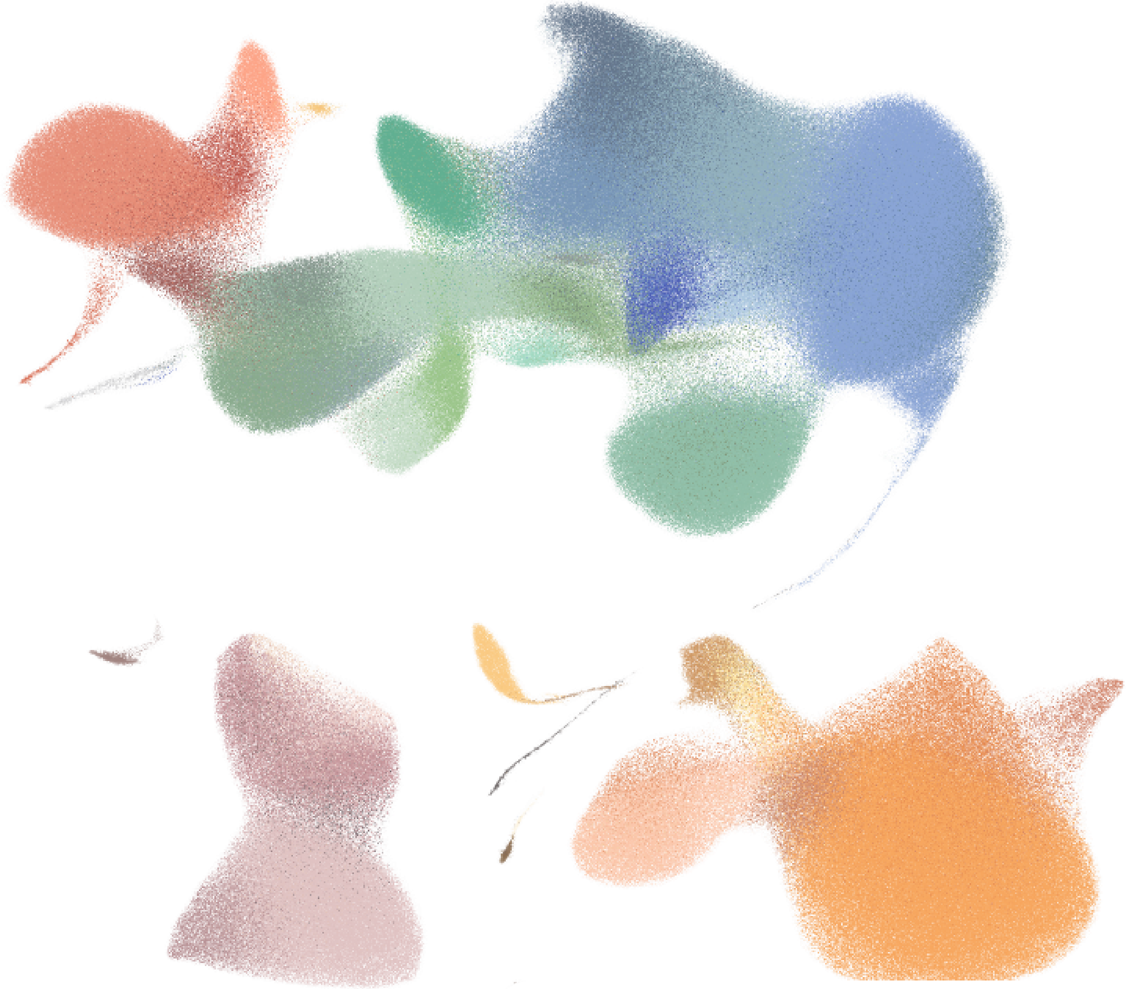


ELIXIR CZECH REPUBLIC

IMG

WHERE IT HAPPENS

Integration merges samples together and corrects the batch effect

01	02	03	04	05
QC & filtering	Normalisation & HVGs	**PCA & integration**	Clustering	Markers & annotation

Integration corrects the embedding

The neighbour graph, clustering and UMAP all read the integrated space. Correcting it once is what fixes every plot that follows.

Do tests on the original data

Differential expression and marker detection stay on the normalised counts, with batch as a covariate. Corrected values inflate significance.

THE PROBLEM

Every dataset carries biological and technical variation at once

Biological variation

What you set out to measure: cell type, treatment response, genotype, disease state, time point.

Keep it.

Technical variation

Everything the protocol adds: chemistry version, library prep, sequencing run, operator, day.

Remove it — if you can tell it apart.

The count matrix does not label which is which. Only the experimental design tells them apart.

WHERE IT COMES FROM

A batch effect appears wherever samples are handled separately

Sample handling

Time to dissociation, enzyme lot, cell viability

Chemistry

10x v2 vs v3, kit lot, capture efficiency

Operator & day

Who ran it, and on which morning

Sequencing

Flow cell, run, depth per cell

Platform

Droplet vs plate-based, 3' vs 5'

Pipeline

Reference genome build, aligner version

CONFOUNDING

If condition and batch coincide, no algorithm can tell them apart

Batch 1 — Monday

control ×4

Batch 2 — Friday

treated ×4

Batch = condition.

Any difference has two explanations; the data cannot choose between them.

Confounded — unrecoverable

Batch 1 — Monday

ctrl ×2 | trt ×2

Batch 2 — Friday

ctrl ×2 | trt ×2

Batch is crossed with condition.

The batch effect can be estimated without touching the biology.

Balanced — correctable

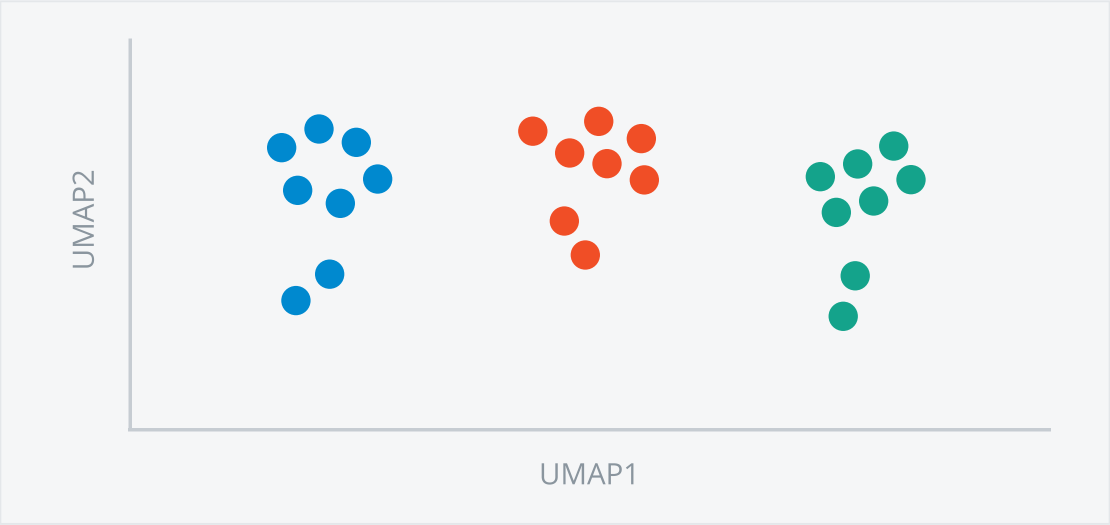
DESIGN FIRST

The cheapest correction is a design that never needed one

Spread conditions	Every batch should contain every condition. Never process all controls on one day.
Multiplex	Hashtags, CellPlex or genotype-based demultiplexing put several samples in one lane — the batch effect disappears instead of being modelled.
Replicate biologically	Several donors or animals per condition. Cells are not replicates; samples are.
Freeze the protocol	One chemistry version, one reference, one pipeline for the whole study.
Record everything	Date, operator, kit lot, run. A batch you did not write down cannot be corrected for.

THE SYMPTOM

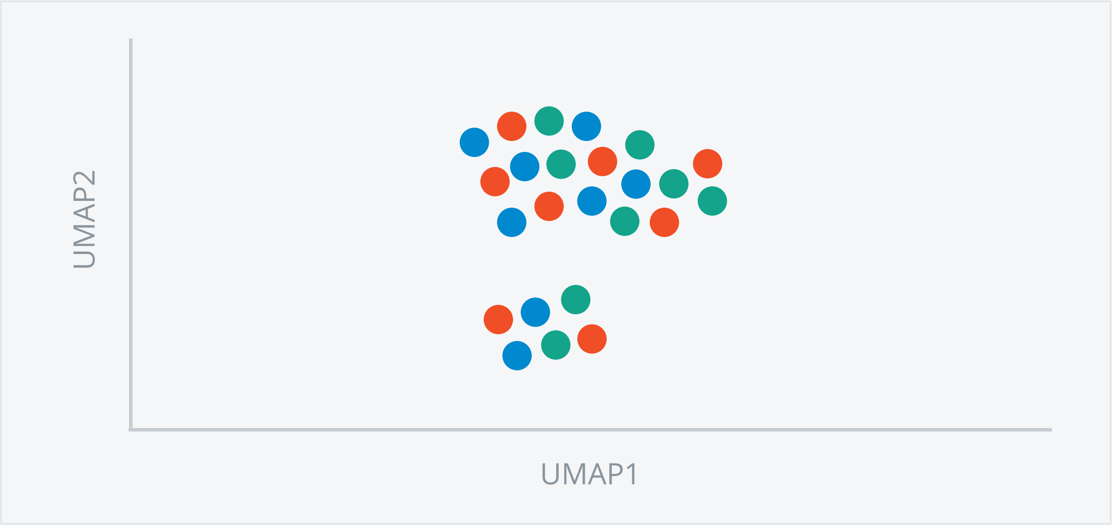
Without correction, cells group by sample instead of by cell type



colour = sample · one cloud per sample

Before integration

Samples sit apart, so clusters track the sample of origin and the shared cell types stay hidden.



two cell-type clusters, samples mixed

After integration

Samples overlap and the real structure appears: two cell-type clusters, one annotation for all.

PART ONE

How the methods work

TAXONOMY

Integration methods differ mainly in what they hand back

Corrected expressions

A new gene × cell matrix with the batch component subtracted.

Seurat CCA v4, fastMNN, ComBat

New joint embedding

Only new low-dimensional coordinates; genes are left untouched.

Harmony, scVI

Integrated graph

Only a neighbour graph in which cells from different batches are linked.

BBKNN, Conos

Differential expression should still be done on the uncorrected counts, with batch in the model.

METHOD — MNN

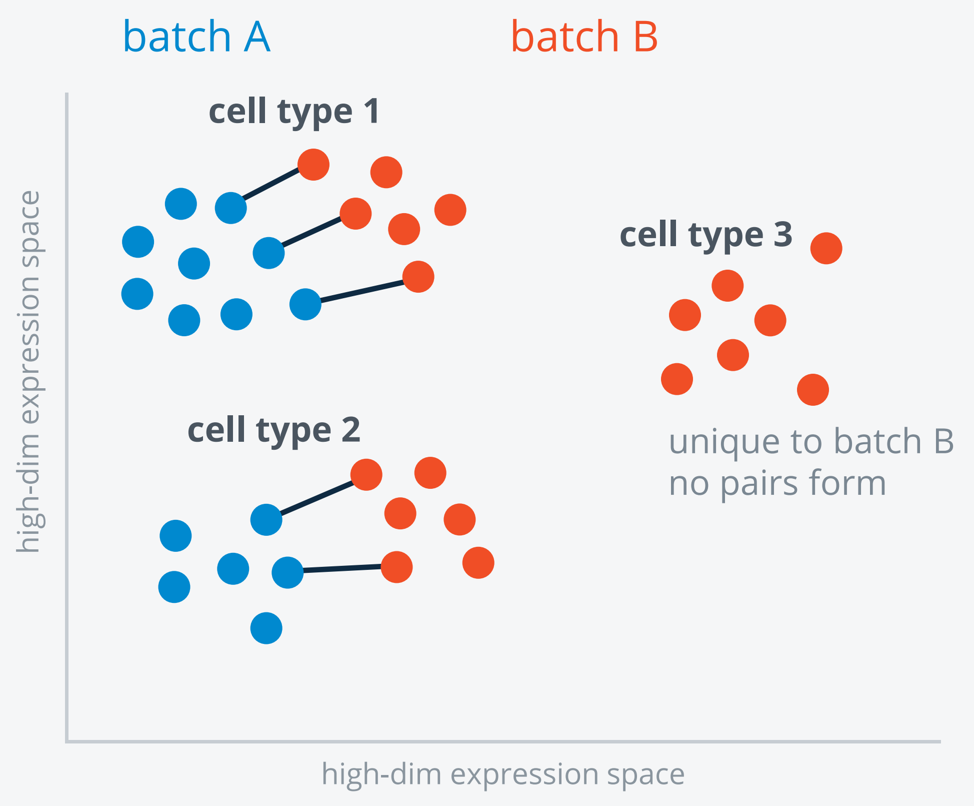
MNN anchors batches on cells that pick each other as neighbours

For every cell in batch A, find its nearest neighbours in batch B, and vice versa. Keep only the pairs that chose *each other*.

Such mutual pairs are very likely to be the same cell type measured twice. The vector between them estimates the batch effect locally.

Correction: smooth those vectors across neighbouring cells and shift one batch onto the other.

Seurat's CCA follows the same logic in a shared correlation space. Assumes the batches share at least one population.



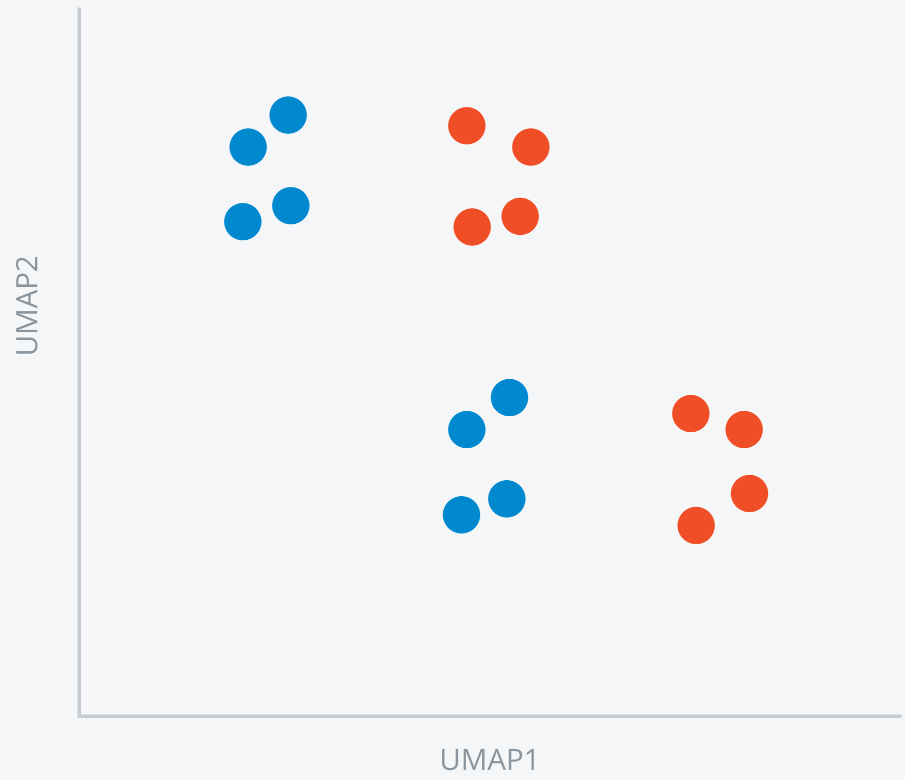
MNN pairs form only where both batches share a population

METHOD — HARMONY

Harmony alternates soft clustering with a per-cluster correction

1. Soft-cluster cells in PCA space, penalising clusters dominated by one batch.
2. In each cluster, compute how far each batch sits from the cluster centre.
3. Shift the cells by that offset, weighted by cluster membership.
4. Repeat until the batches stop moving.

Very fast, scales to millions of cells, handles several covariates at once. Returns an embedding only — no corrected gene values.



colour = batch · batches sit apart inside shared populations

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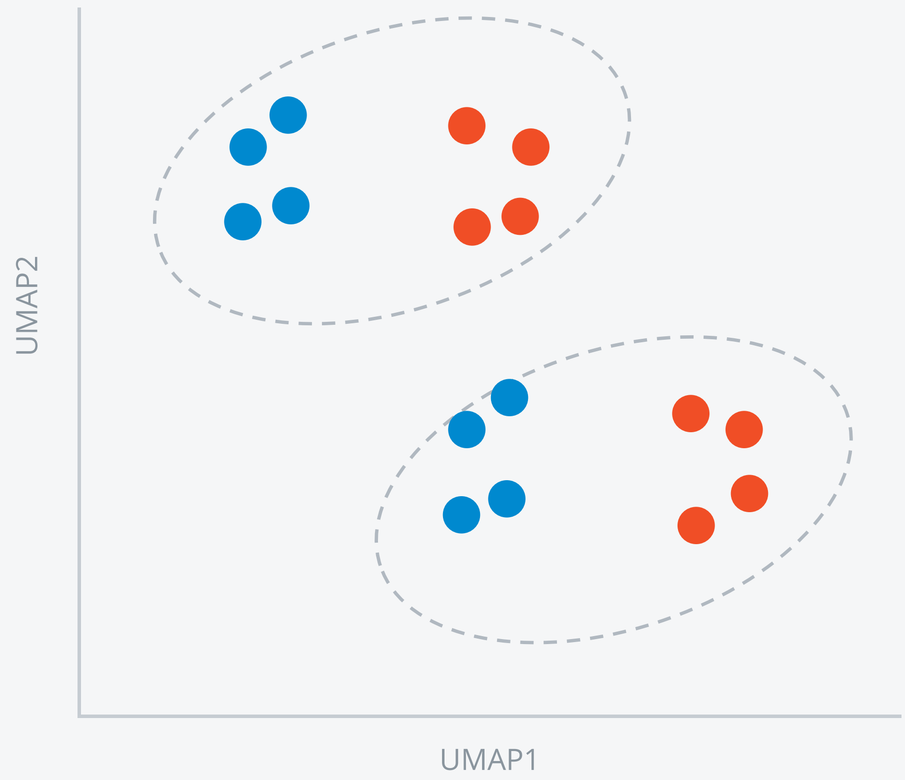

soft clusters found · each cell gets a membership weight

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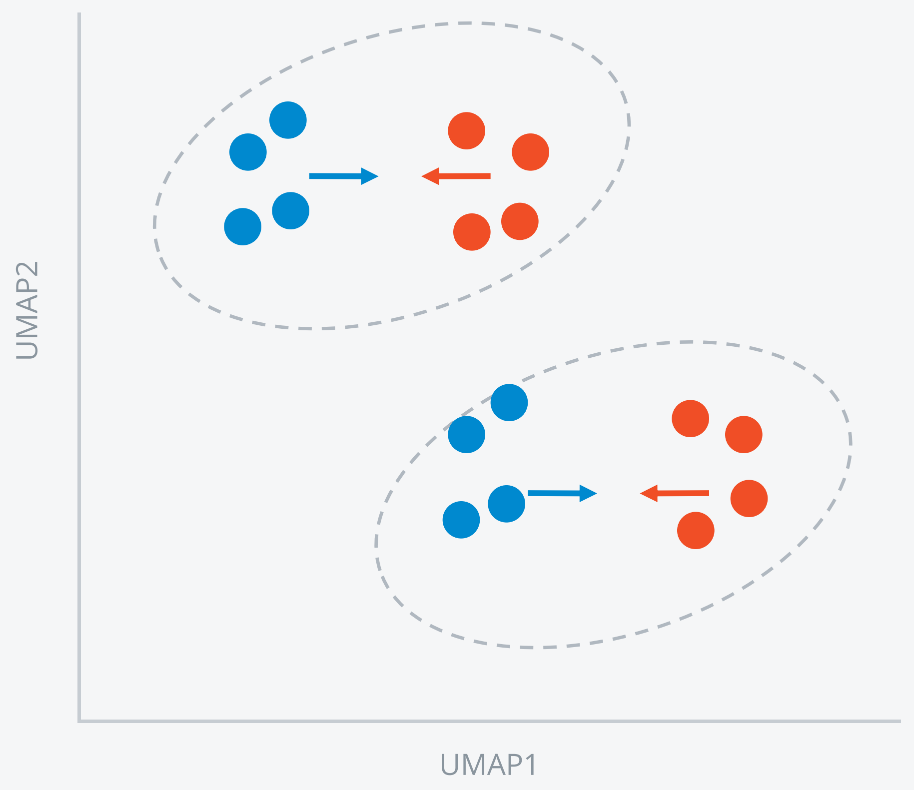

each batch offset from its cluster centre is measured

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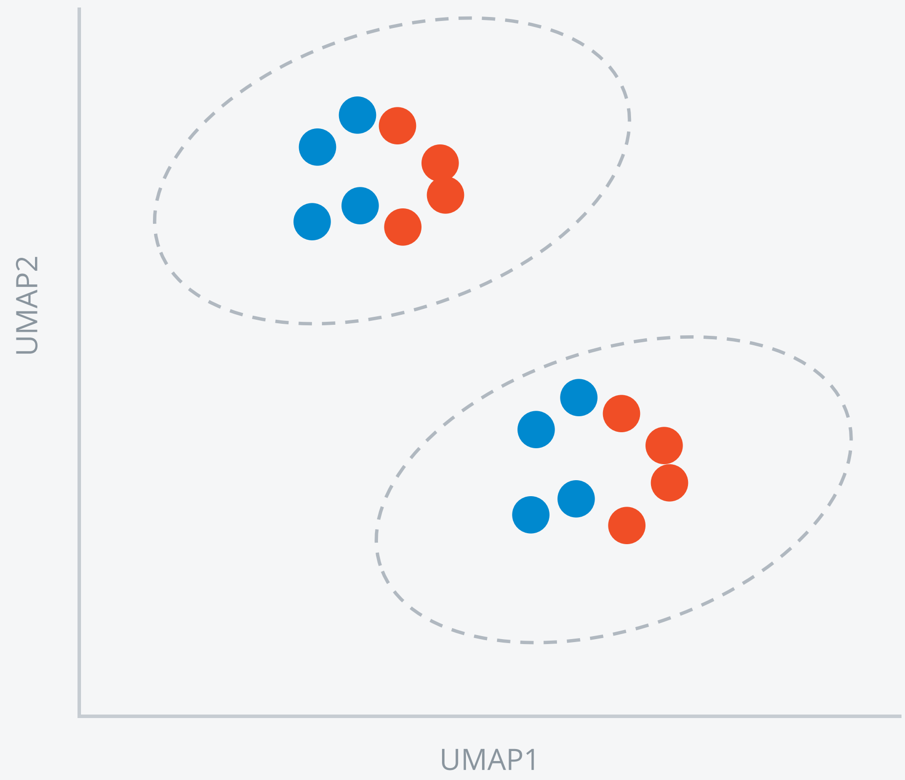

cells move by their cluster offset · batches start to overlap

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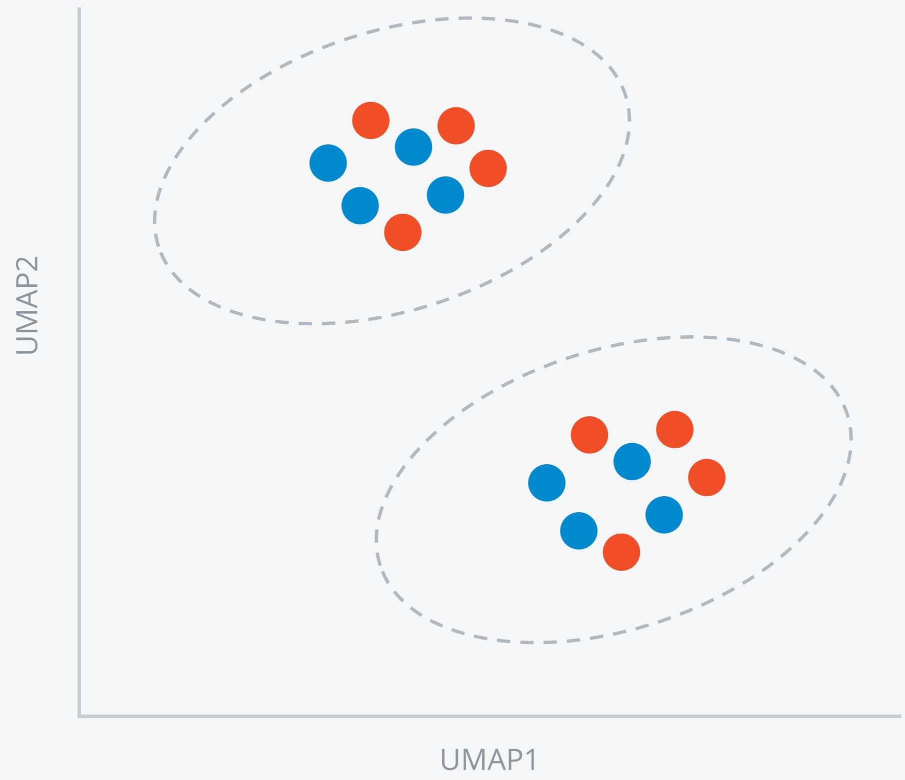


converged · batches mixed inside both clusters, structure kept

scVI learns a latent space without the batch label

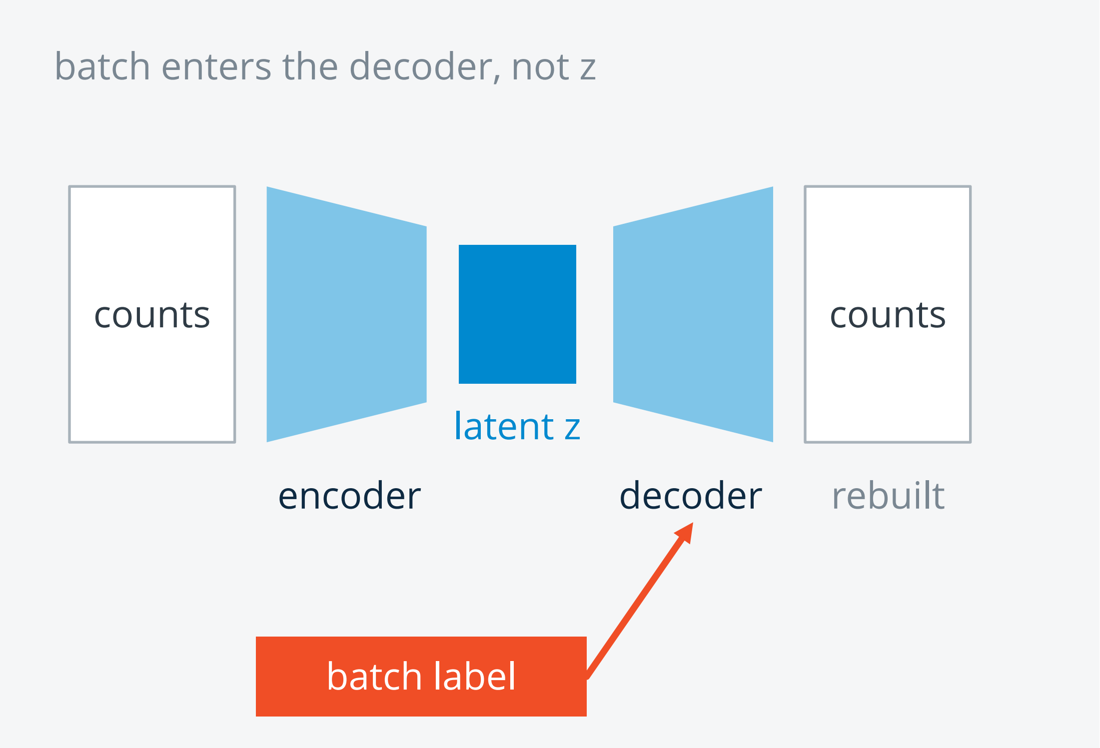
A neural network compresses each cell's counts into ~10–30 latent dimensions, then reconstructs the counts from them — while also receiving the batch label as an input.

Because batch is supplied separately, the latent space has no reason to encode it. What remains is cell state.

Strong with many batches, deep hierarchies and large atlases; models counts directly.

Costs GPU time, hyperparameters, and a stochastic result — set the seed.

scANVI extends it with known labels for some cells.



PART TWO

Did it work — and should you have done it?

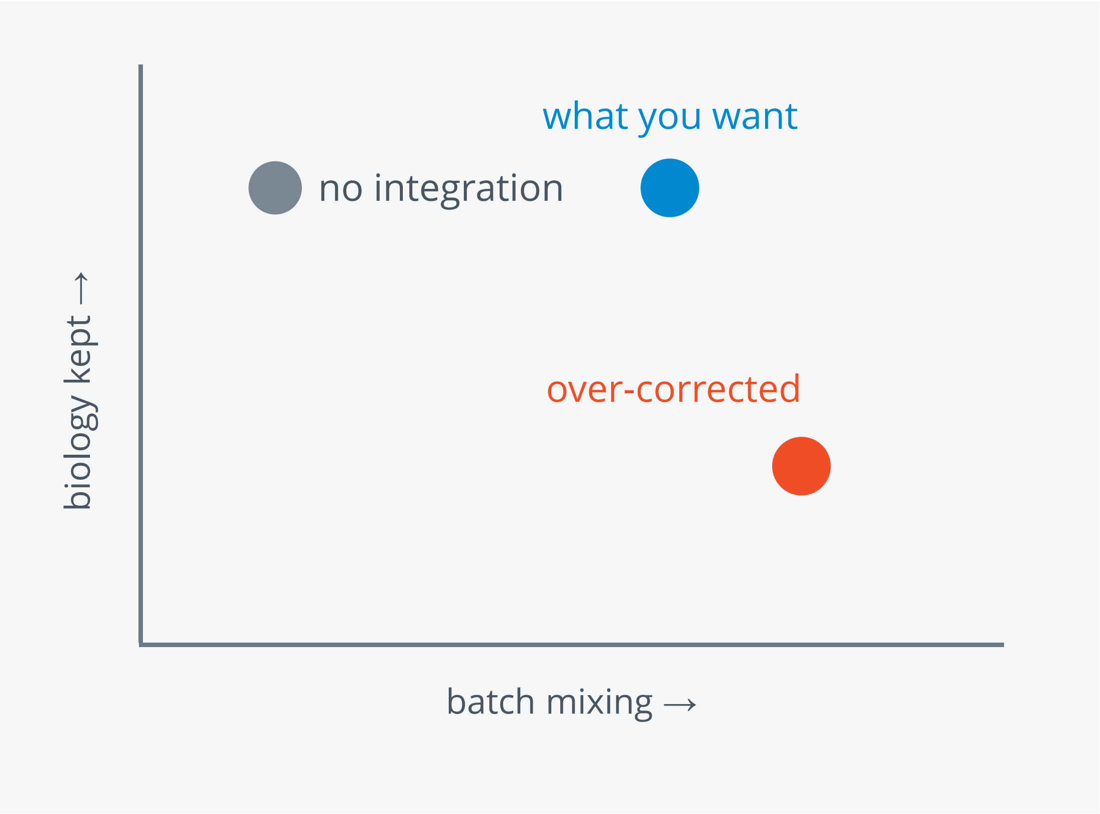
THE TRADE-OFF

Good integration mixes batches without erasing cell types

Every parameter that improves batch mixing also pushes distinct populations together. The two goals pull in opposite directions.

Over-correction is the quiet failure: the UMAP looks beautifully mixed, and a rare population present in only one sample has been smeared into its neighbour.

Always compare the corrected result with the uncorrected one before trusting it.



EVALUATION

Judge integration on both axes, never on the UMAP alone

Batch mixing

Each cell's neighbourhood should carry the batch composition of the whole dataset.

The effective number of batches around a cell should be high — higher means better mixed.

Batch separation should sit near zero — batches must not form their own clusters.

Biology kept

Clusters after integration should still match the known cell-type labels.

Cell-type separation should stay high — populations must remain separable.

Known markers must still mark the same cells, and rare populations must survive.

Do not integrate when the batch *is* the biology

Tumour and healthy tissue, or two species — the populations genuinely differ; forcing them to overlap invents shared cell types.

One sample per condition — correcting the batch removes the condition with it.

Differential expression testing — model batch as a covariate on raw counts instead.

WATCH OUT

Most integration problems come from a handful of habits

01 Treating a well-mixed UMAP as proof that integration succeeded.

02 Integrating before QC, so dying cells and doublets are aligned across batches.

03 Running differential expression on the corrected matrix instead of the counts.

04 Declaring the wrong batch variable — donor when the real split was the sequencing run.

05 Selecting variable genes on all samples at once, so batch-driven genes lead the PCA.

06 Never looking at the uncorrected embedding, so the size of the effect stays unknown.

CHEAT SHEET

Integration, in the order you will actually do it

1. Write down the design: which variable is biology, which is batch, are they crossed?
2. QC and normalise each sample first; select variable genes per batch, then combine.
3. Run PCA and UMAP *without* correction — measure how bad the batch effect is.
4. If it dominates, integrate — start with Harmony, escalate if it under-corrects.
5. Cluster on the integrated embedding; colour by batch, cluster and markers.
6. Check both axes: batches mixed, cell types still separable, rare populations intact.
7. Keep the uncorrected counts for differential-expression test.
8. Report the method, the batch variable and the seed.

FURTHER READING

Where to go next

Single-cell best practices — integration chapter · sc-best-practices.org

Seurat introduction to scRNA-seq integration · satijalab.org/seurat

Luecken et al. (2022), *Benchmarking atlas-level data integration* · Nature Methods

Haghverdi et al. (2018), MNN correction · Nature Biotechnology

Korsunsky et al. (2019), Harmony · Nature Methods · Lopez et al. (2018), scVI · Nature Methods

Thank you

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